

Job Portal System

Requirement Analysis:

Project Overview:

The Job Portal System serves as a platform for job seekers and employers to connect and manage job applications. Job seekers can search and apply for jobs, while employers can post job openings, track applications, and manage the hiring process.

Entities

1. JobSeeker

- **Description:** Represents individuals seeking employment.
- **Attributes:** jobSeekerId, firstName, lastName, email, phone, address, dateOfBirth, profilePicture, createdAt, updatedAt.

2. Employer

- **Description:** Represents organizations or individuals posting job openings.
- **Attributes:** employerId, companyName, contactPerson, email, phone, address, createdAt, updatedAt.

3. JobPost

- **Description:** Represents job listings posted by employers.
- **Attributes:** jobPostId, employerId, title, description, categoryId, jobTypeId, location, salary, datePosted, status, createdAt, updatedAt.

4. Application

- **Description:** Represents job applications submitted by job seekers.
- **Attributes:** applicationId, jobPostId, jobSeekerId, resumeId, status, dateApplied, createdAt, updatedAt.

5. Resume

- **Description:** Represents job seeker's resumes.
- **Attributes:** resumeId, jobSeekerId, filePath, createdAt, updatedAt.

6. Interview

- **Description:** Represents interviews scheduled for applicants.
- **Attributes:** interviewId, applicationId, date, time, location, status, createdAt, updatedAt.

7. Notification

- **Description:** Represents alerts sent to users regarding job postings and application updates.
- **Attributes:** notificationId, userId, message, isRead, createdAt.

8. Category

- **Description:** Represents different job categories (e.g., IT, Finance).

- **Attributes:** categoryId, name, createdAt, updatedAt.

9. Skill

- **Description:** Represents skills associated with job seekers and job postings.
- **Attributes:** skillId, name, createdAt, updatedAt.

10. Payment

- **Description:** Represents payment transactions for job postings or subscriptions.
- **Attributes:** paymentId, userId, amount, date, status, createdAt.

11. Report

- **Description:** Represents reports on job application statistics and hiring trends.
- **Attributes:** reportId, employerId, content, generatedAt.

12. Feedback

- **Description:** Represents feedback provided by job seekers about the job portal.
- **Attributes:** feedbackId, userId, message, createdAt.

13. JobType

- **Description:** Represents types of jobs (e.g., Full-time, Part-time, Freelance).
- **Attributes:** jobTypeId, name, createdAt, updatedAt.

14. Offer

- **Description:** Represents job offers extended to job seekers.
- **Attributes:** offerId, applicationId, salary, startDate, status, createdAt.

15. Role

- **Description:** Represents roles assigned to users (e.g., Job Seeker, Employer).
- **Attributes:** roleId, name, createdAt, updatedAt.

Modules

1. Job Posting

- Employers can create, edit, and manage job postings.
- Features: Upload job details, set application deadlines, and manage job statuses.

2. Resume Management

- Job seekers can upload and manage their resumes.
- Features: Create multiple resume versions, edit, and delete resumes.

3. Application Tracking

- Employers can view and manage job applications.
- Features: Filter applications by status, view applicant details, and manage interview scheduling.

4. Interview Scheduling

- Manage and schedule interviews for applicants.

- Features: Set interview dates and times, send notifications to applicants.

5. **Payment Gateway**

- Manage payments for premium job postings and subscriptions.
- Features: Process transactions, handle refunds, and generate payment reports.

6. **Notification System**

- Alerts for new job postings and application status updates.
- Features: Send notifications via email or in-app alerts.

7. **Reporting Module**

- Generate reports on hiring trends and application statistics.
- Features: Custom report generation based on various metrics.

Workflow

1. **Job Seeker Flow:**

- Register/Login.
- Create and upload resume.
- Browse job postings by category or skill.
- Apply for jobs by submitting applications and resumes.
- Receive notifications about application status and interview schedules.

2. **Employer Flow:**

- Register/Login.
- Post new job listings.
- Review applications and track their status.
- Schedule interviews with selected candidates.
- Make job offers to selected candidates.

3. **Payment Flow:**

- Employers can choose to post jobs for free or opt for premium features.
- Payment processing through a secure gateway.
- Generation of payment receipts and invoices.

4. **Notification Flow:**

- Users receive notifications for new job postings, application updates, and interview schedules.
- Notifications are accessible in-app and can be sent via email.

Conclusion

This requirement analysis for the **Job Portal System** outlines the essential components and functionalities needed to create a robust platform for job seekers and employers. With clearly defined entities and modules, you

can proceed to design the database structure and develop the application using Core Java, Collections, and Hibernate as the ORM tool.

Detailed Modules Description

1. Job Posting Module

- **Purpose:** To enable employers to create, manage, and update job listings.
 - **Key Features:**
 - **Create Job Post:** Employers can input job details such as title, description, requirements, salary, job type, and category.
 - **Edit Job Post:** Employers can modify existing job listings.
 - **Delete Job Post:** Employers can remove job postings when they are no longer needed.
 - **Manage Applications:** View the number of applications received for each job post.
 - **Job Status Management:** Set the job status (open, closed, filled) and update it as necessary.
 - **Job Expiry Notification:** Notify employers when their job posts are about to expire.
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2. Resume Management Module

- **Purpose:** To facilitate job seekers in managing their resumes effectively.
 - **Key Features:**
 - **Upload Resume:** Allow job seekers to upload their resumes in various formats (PDF, DOCX).
 - **Create Multiple Resumes:** Enable users to create and maintain multiple resume versions for different job applications.
 - **Edit Resume:** Allow users to modify their existing resumes easily.
 - **Delete Resume:** Users can remove resumes that they no longer wish to keep.
 - **Preview Resume:** Provide a preview option to view how the resume looks before submission.
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3. Application Tracking Module

- **Purpose:** To allow employers to manage and track job applications received for their postings.
 - **Key Features:**
 - **View Applications:** Employers can see all applications for a specific job post.
 - **Filter Applications:** Sort applications based on status (new, under review, interviewed, hired, rejected).
 - **Application Status Update:** Employers can update the status of applications based on the hiring process.
 - **Candidate Profiles:** View detailed candidate profiles, including resumes, skills, and cover letters.
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4. Interview Scheduling Module

- **Purpose:** To manage the interview process for job applicants.
- **Key Features:**

- **Schedule Interviews:** Employers can set date and time for interviews and invite candidates.
 - **Manage Interview Rounds:** Define multiple interview rounds and schedule them accordingly.
 - **Send Notifications:** Automatically notify candidates about interview details through email or app notifications.
 - **Track Interview Feedback:** Employers can provide feedback after interviews and track candidate performance.
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5. Payment Gateway Module

- **Purpose:** To handle all payment-related functionalities for employers and job seekers.
 - **Key Features:**
 - **Payment Processing:** Securely process payments for premium job postings and job seeker subscriptions.
 - **Manage Transactions:** Keep records of all payment transactions, including amounts and dates.
 - **Refund Management:** Handle refund requests and processes.
 - **Payment Reports:** Generate reports for payment activities, including income from job postings.
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6. Notification System Module

- **Purpose:** To inform users about important events and updates within the system.
 - **Key Features:**
 - **Real-Time Notifications:** Notify users of new job postings, application status changes, and interview schedules.
 - **Email Notifications:** Send email alerts for critical updates, ensuring users are informed even when not logged in.
 - **In-App Notifications:** Provide an in-app notifications section for easy access to alerts.
 - **Notification Preferences:** Allow users to customize their notification preferences (e.g., frequency, types of alerts).
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7. Reporting Module

- **Purpose:** To generate insights and reports for both employers and administrators.
 - **Key Features:**
 - **Hiring Trends Reports:** Generate reports based on hiring trends over time, showcasing successful job posts.
 - **Job Application Statistics:** Provide detailed statistics on the number of applications received, accepted, and rejected.
 - **User Activity Reports:** Track user engagement with the platform (job seekers and employers).
 - **Export Reports:** Allow users to export reports in various formats (CSV, PDF) for offline analysis.
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Conclusion

These modules collectively create a comprehensive **Job Portal System** that connects job seekers and employers while managing the complexities of job applications and postings. Each module has specific functionalities that contribute to a seamless user experience, making the platform efficient and user-friendly.

Detailed System Workflow

1. User Registration and Profile Setup

- **Job Seeker Registration:**

1. User visits the job portal.
2. Selects "Sign Up" and fills in the registration form (name, email, password).
3. Confirms email address through a verification link.
4. Completes profile setup by adding personal details, skills, and uploading a resume.

- **Employer Registration:**

1. User visits the job portal.
 2. Selects "Employer Sign Up" and fills in the registration form (company name, email, password).
 3. Confirms email address through a verification link.
 4. Completes profile setup by providing company details and logo.
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2. Job Posting by Employers

1. Employer logs into the portal.
 2. Navigates to the "Job Posting" module.
 3. Fills in job details:
 - Job title
 - Description
 - Requirements
 - Category
 - Salary range
 - Job type (full-time, part-time, freelance)
 4. Submits the job post for review or publishes it immediately.
 5. Receives confirmation of the job post being live on the portal.
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3. Job Seeker Job Search and Application

1. Job seeker logs into the portal.
2. Navigates to the "Job Search" section.
3. Uses filters (category, location, job type) to search for jobs.
4. Views job details and decides to apply for a selected job.
5. Clicks "Apply Now" and submits an application, which may include:
 - Resume (uploaded earlier or uploaded anew)

- Cover letter (optional)
- Additional information (questions specific to the job post)

6. Receives confirmation of the application submission.

4. Application Tracking by Employers

1. Employer logs into the portal.
 2. Navigates to the "Application Tracking" module.
 3. Views all applications for the posted jobs.
 4. Filters applications based on status (new, under review, interviewed).
 5. Updates application status as needed (e.g., selects candidates for interviews).
 6. Sends notifications to candidates about their application status via email or in-app notifications.
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5. Interview Scheduling

1. Employer selects a candidate for an interview.
 2. Navigates to the "Interview Scheduling" module.
 3. Chooses the interview date and time, and selects the interview format (in-person, virtual).
 4. Sends an interview invitation to the candidate.
 5. Candidate receives the interview notification and confirms their availability.
 6. Both parties can access the scheduled interview details in their respective accounts.
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6. Feedback and Decision Making

1. After the interview, the employer provides feedback on the candidate.
 2. Updates the application status to "hired" or "rejected."
 3. If hired, the employer can send a job offer to the candidate.
 4. Candidate accepts or declines the offer, and both parties receive notifications.
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7. Payment Processing

1. For premium job postings, the employer navigates to the "Payment Gateway" module.
 2. Selects the desired payment plan (e.g., one-time, subscription).
 3. Enters payment details and submits the payment.
 4. Receives confirmation of the payment and generates an invoice.
 5. The system records the transaction for future reference.
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8. Notification System

1. Users receive real-time notifications for:

- New job postings matching their profile
 - Updates on application statuses
 - Interview schedules and feedback
2. Users can manage their notification preferences (opt-in/opt-out).
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9. Reporting Module

1. Administrators or employers can access the "Reporting Module."
 2. Generates reports on:
 - Hiring trends (number of jobs posted, filled, and rejected)
 - Application statistics (applications per job, success rates)
 - User engagement metrics (job seeker activities)
 3. Users can export reports in CSV or PDF formats for offline analysis.
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Conclusion

This workflow provides a comprehensive overview of how users interact with the **Job Portal System**, highlighting the flow of actions from registration to job application tracking, interview scheduling, and payment processing. Each step ensures a seamless experience for both job seekers and employers, promoting effective communication and management of the hiring process.